◆ WEEK 2 — SESSION 6

DATABASES

Amazon RDS vs DynamoDB

SESSION NOTES

◆ SESSION OBJECTIVE

By the end of this session, you should understand:

✓ What databases are
✓ Why databases matter in cloud computing
✓ The difference between relational and NoSQL databases
✓ How Amazon RDS works
✓ How Amazon DynamoDB works
✓ When to choose the correct database architecture

◆ WHAT IS A DATABASE?

A database is a system used to:

- Store information
- Organize information
- Retrieve information
- Manage information efficiently

Databases allow applications to remember data.

Examples:

- Banking systems
- Hospital systems
- Social media platforms
- E-commerce websites
- Mobile applications

◆ WHY DATABASES MATTER

Modern applications depend on databases.

Without databases:

✕ Websites cannot store users
✕ Applications cannot save transactions
✕ Businesses cannot manage information
✕ Systems cannot scale properly

Databases are part of the core infrastructure of modern cloud systems.

◆ REAL-WORLD EXAMPLES

Business System	Stored Data
Banking App	Transactions, balances
Hospital System	Patient records
E-commerce Platform	Products, customers
Social Media App	Posts, likes, comments
School System	Students, marks

◆ DATABASE CATEGORIES

There are two major database models:

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1. RELATIONAL DATABASES

Relational databases store data in:

✓ tables
✓ rows
✓ columns

Data is highly structured and organized.

Examples:

ID	Name	Email
1	Sarah	sarah@email.com

◆ CHARACTERISTICS OF RELATIONAL DATABASES

Relational databases:

✓ use structured schemas
✓ maintain relationships between data
✓ support SQL queries
✓ prioritize consistency and accuracy

◆ RELATIONAL DATABASE USE CASES

Best for:

- Banking systems
- Financial systems
- Hospital records
- ERP systems
- Inventory management

These systems require:

✓ accuracy
✓ consistency
✓ structured relationships

◆ WHAT IS AMAZON RDS?

Amazon RDS stands for:

Relational Database Service

It is AWS's managed relational database platform.

RDS allows organizations to run databases without managing:

- physical servers
- operating systems
- hardware maintenance

AWS handles much of the infrastructure automatically.

◆ WHAT AWS MANAGES IN RDS

AWS manages:

✓ backups
✓ patching
✓ availability
✓ infrastructure
✓ monitoring
✓ scaling support

This reduces operational complexity.

◆ SUPPORTED RDS DATABASE ENGINES

Amazon RDS supports:

- MySQL
- PostgreSQL
- MariaDB
- Oracle
- Microsoft SQL Server

◆ WHY BUSINESSES USE RDS

Businesses use RDS because it provides:

✓ reliability
✓ managed operations
✓ scalability
✓ enterprise-grade database support

◆ SIMPLE ANALOGY — RELATIONAL DATABASES

Think of relational databases like a highly organized filing cabinet.

Every document has:

✓ a specific folder
✓ a specific structure
✓ defined relationships

Nothing is randomly placed.

This structure improves consistency and control.

◆ WHAT IS NOSQL?

NoSQL databases are designed for:

✓ flexibility
✓ speed
✓ massive scale
✓ cloud-native applications

Unlike relational databases, NoSQL databases do not require strict table structures.

◆ CHARACTERISTICS OF NOSQL DATABASES

NoSQL systems are:

✓ flexible
✓ horizontally scalable
✓ optimized for large-scale workloads
✓ designed for cloud-native environments

◆ WHAT IS AMAZON DYNAMODB?

Amazon DynamoDB is AWS's fully managed NoSQL database service.

It is designed for:

✓ ultra-fast performance
✓ automatic scaling
✓ serverless applications
✓ massive workloads

◆ WHY DYNAMODB IS POWERFUL

DynamoDB provides:

✓ low latency
✓ automatic scaling
✓ serverless architecture
✓ high performance at scale

It is commonly used for:

- gaming systems
- mobile applications
- IoT platforms
- real-time systems

◆ WHAT "SERVERLESS DATABASE" MEANS

Serverless does NOT mean there are no servers.

It means:

AWS manages the database infrastructure automatically.

You focus on:

✓ applications
✓ data
✓ development

AWS handles:

✓ scaling
✓ infrastructure
✓ maintenance
✓ availability

◆ SIMPLE ANALOGY — DYNAMODB

Think of DynamoDB like a massive smart warehouse.

Instead of placing items into fixed shelves and folders:

The system automatically organizes and scales storage dynamically based on demand.

This makes it fast and highly scalable.

◆ RDS VS DYNAMODB

Feature	Amazon RDS	DynamoDB
Database Type	Relational	NoSQL
Structure	Tables & relationships	Flexible key-value
Scaling	Traditional scaling	Automatic scaling
Best For	Structured business systems	High-scale applications
Query Language	SQL	API-based access
Server Management	Managed	Fully serverless

◆ WHEN TO USE AMAZON RDS

Use RDS when:

✓ relationships between data matter
✓ transactions must remain accurate
✓ systems require structured schemas
✓ consistency is critical

Examples:

- banking
- payroll
- ERP systems
- healthcare records

◆ WHEN TO USE DYNAMODB

Use DynamoDB when:

✓ workloads scale rapidly
✓ applications are event-driven
✓ flexibility is important
✓ ultra-fast performance is required

Examples:

- gaming
- mobile apps
- IoT systems
- social platforms

◆ IMPORTANT ARCHITECTURE THINKING

Cloud architects do NOT select services randomly.

They evaluate:

✓ business requirements
✓ scalability needs
✓ cost considerations
✓ performance expectations
✓ operational complexity

◆ WHY DATABASE DECISIONS MATTER

Choosing the wrong database can cause:

✕ poor performance
✕ high costs
✕ scaling failures
✕ operational complexity
✕ architecture instability

Correct database selection is part of cloud governance and architecture strategy.

◆ SECURITY CONSIDERATIONS

Databases contain critical information.

Cloud engineers must think about:

✓ encryption
✓ access control
✓ backups
✓ monitoring
✓ disaster recovery

Weak database security can expose entire organizations.

◆ KEY TAKEAWAYS

✓ Databases store and organize information
✓ Relational databases use structured tables

✓ NoSQL databases focus on scalability and flexibility
✓ Amazon RDS is AWS's managed relational database service
✓ DynamoDB is AWS's serverless NoSQL database
✓ Database decisions affect performance, scalability, and cost

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◆ QUICK RECALL QUESTIONS

1. What is a database?
2. What is the difference between relational and NoSQL databases?
3. What does Amazon RDS provide?
4. Why is DynamoDB considered serverless?
5. When should architects use relational databases?
6. Why is scalability important in cloud systems?

◆ FINAL THOUGHT

"Modern cloud architecture is not about using the most powerful service.

It is about selecting the correct service for the correct workload."